

MATH 2130 - Week 2

Vector Spaces, Subspaces, and Linear Transformations

1 Overview

This week we will generalize and abstract the ideas that we developed in week 1. We will consider (abstract) **vector spaces**. We will look at **subspaces** of vector spaces and **linear transformations** between them. Building off of what we did in week 1, we will look at systems of m equations in n unknowns and discuss their solutions (or lack thereof). This will lead to **basic variables**, **free variables** and **pivots** (which we have already seen). The most important subspaces that we will look at are the **nullspace** and **column space** of a linear transformation. This will lead to the idea of the **rank** of a linear transformation.

While we will work with abstract vector spaces and linear transformations between them, we will also make things more concrete by talking about the **matrix representation** of a linear transformation. To do this, we will need to talk about **linear independence** and **spanning sets** of vectors. These ideas will combine in the idea of a **basis** for a vector space and the associated matrix representation. We will also talk about the **rank** of a matrix A .

Connected to any linear transformation there are **Four Fundamental Subspaces**. We will look at these and their interrelations. A natural idea that emerges from this is **orthogonality**, which generalizes the idea of perpendicularity to abstract vector spaces. This topic will be covered in more depth during week 3.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) Nonsingular Matrices and Inverses. Summary of Solving Systems of Linear Equations. Vector Spaces and Subspaces. Linear Transformations.
- (T) Systems of m equations in n unknowns. Basic Variables and Free Variables.
- (W) Linear Independence, Basis, and Dimension.
- (R) Nullspace and Column Space. The Four Fundamental Subspaces.
- (F) Orthogonality of Vectors and Subspaces.

Here is a checklist of things to do this week. This is meant to help you keep track of what you've completed and what you still need to work on.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Austin and Olver.
- ☐ Complete all of the homework exercises.

3 Key Concepts

- Vector Spaces and Subspaces
- Linear Transformations
- Systems of m equations in n unknowns
- Linear Independence and Span
- Basis and Dimension
- Rank of a Linear Transformation/Matrix
- Nullspace (Kernel) and Column Space
- The Four Fundamental Subspaces
- Orthogonal Vectors and Subspaces

4 Important Definitions, Formulas, and Theorems

- Let F be a field (usually $\mathbb{R}$ or $\mathbb{C}$). A **vector space** is a set, V , of objects, called **vectors**, that is closed under the following two operations:
 - If $x, y \in V$, then $x + y \in V$.
 - If $x \in V$ and c is any scalar in F , then $cx \in V$.
- The Subspace Test: A subset S of a vector space V is a subspace if it is closed under addition and scalar multiplication:
 - If $x, y \in S$, then $x + y \in S$.
 - If $x \in S$ and c is any scalar, then $cx \in S$.
- Let A be an $m \times n$ matrix with all entries in $\mathbb{R}$. The system $Ax = b$ is solvable if and only if the vector b can be expressed as a linear combination of the columns of A . That is, if $a_1, a_2, \dots, a_n$ are the columns of A , then we can find scalars $x_1, x_2, \dots, x_n$, called **weights**, such that

$$x_1 a_1 + x_2 a_2 + \dots + x_n a_n = b$$

- The subspace formed by all linear combinations of the columns of A is called the **column space** of the matrix A . It is a subspace of $\mathbb{R}^m$.
- The set of all solutions to $Ax = \vec{0}$ is called the **nullspace** of the matrix A . It is a subspace of $\mathbb{R}^n$.
- Given any $m \times n$ matrix A , we can always transform A into **echelon form** by a sequence of elementary row operations. The echelon form has the three following properties:

1. The nonzero rows come first. The first nonzero entry in these nonzero rows are called the **pivots**.
2. Below each pivot is a column of zeros.
3. Each pivot lies to the right of the pivot in the row above it.

In general, a matrix A in echelon form, call it U , will look like the following:

$$U = \begin{bmatrix} p_1 & * & * & * & * & * \\ 0 & p_2 & * & * & * & * \\ 0 & 0 & 0 & p_3 & * & * \\ 0 & 0 & 0 & 0 & 0 & p_4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The p_i are the pivots.

- Let A be an $m \times n$ matrix. Put it into echelon form, U . Let the vector x be the n -dimensional vector of variables

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The **basic variables** are the entries of x that correspond to the columns of U with pivots. All the other entries of x are called **free variables**. They correspond to the columns of U without pivots.

In the example for U given above, x would be 6-dimensional and the basic variables are x_1, x_2, x_4, x_6 and the free variables are x_3, x_5 .

- If a matrix A has more columns than rows, so $n > m$, then the **homogeneous equation** $Ax = \vec{0}$ has a nontrivial solution. This means that the nullspace of A contains nonzero vectors. In fact, the “dimension” of the nullspace is equal to the number of free variables (which is the same as the number of columns without pivots in the echelon form of A).
- Suppose the $m \times n$ matrix A is reduced to echelon form U . Let U have r nonzero pivots. This means that the last $m - r$ rows of U are zero. Then there will be r basic variables and $n - r$ free variables. The nullspace, formed of solutions to $Ax = \vec{0}$, has the $n - r$ free variables as parameters. If $r = n$, then there are no free variables and the nullspace contains only the zero vector $x = \vec{0}$.

Solutions to $Ax = b$ exist for every b if and only if $r = m$. In this case, U has no zero rows and every system $Ux = c$ can be solved by back substitution. If $r < m$, then U will have $m - r$ zero rows and there are $m - r$ constraints on b in order for $Ax = b$ to be solvable. If one particular solution exists, then every other solution differs from it by a vector in the nullspace of A .

The number r is called the **rank** of the matrix A .

- Let $v_1, v_2, \dots, v_k$ be a given set of vectors. If all nontrivial combinations of the vectors are nonzero, that is

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k \neq 0 \quad \text{unless} \quad c_i = 0 \quad \text{for all } i$$

then the vectors $v_1, v_2, \dots, v_k$ are **linearly independent**. Otherwise they are **linearly dependent** and at least one of the v_i can be written as a linear combination of the other v 's.

A set of linearly independent vectors will sometimes be called a **linearly independent set**.

- The r nonzero rows of an echelon matrix U are linearly independent, and so are the r columns that contain pivots.
- A set of k vectors in $\mathbb{R}^m$ must be linearly dependent if $k > m$.
- If a vector space V consists of all linear combinations of the particular vectors $w_1, w_2, \dots, w_k$, then these vectors **span** the vector space V . In other words, every v in V can be written as a linear combination of the w 's:

$$v = c_1 w_1 + c_2 w_2 + \dots + c_k w_k$$

If a set of vectors span V , it will sometimes be called a **spanning set**.

- A **basis** for the vector space V is a set of vectors having the following two properties:
 1. It is linearly independent.
 2. It spans V .

We will usually denote a basis by $B = \{b_1, b_2, \dots, b_n\}$.

- A basis for a vector space is **NOT** unique.
- Any two bases for a vector space V contain the same number of vectors. This number, which is shared by all bases and expresses the number of “degrees of freedom” of the space, is called the **dimension** of V .
- Any linearly independent set of vectors in V can be extended to a basis for V , by adding vectors if necessary.

Any spanning set of vectors in V can be reduced to a basis for V , by discarding vectors if necessary.

- The vector space spanned by the rows of a matrix A is called the **row space** of A .
- The row space of A has the same dimension as the row space of U , the echelon form of A . If U has rank r , then the dimension of the row space of both U and A is r . Whats more, if B is a basis for the row space of U then B is also a basis for the row space of A .
- Let A be an $m \times n$ matrix of rank r . The nullspace of A , denoted $N(A)$, has dimension $n - r$. The nullspace of A is sometimes also called the **kernel** of A . The dimension of the nullspace is called the **nullity**, which in this case will be $n - r$.
- The column space of A is sometimes also called the **range** of A . For this reason, the column space of A is denoted by $R(A)$. Just to make things even more confusing, it is also sometimes called the **image** of A .
- The **transpose** of an $m \times n$ matrix A is the $n \times m$ matrix A^T that has the same rows as the columns of A . So to get A^T from A , we switch rows and columns. This is equivalent to saying that $(A^T)_{ij} = (A)_{ji}$. In words, the (ij) -entry of A^T is the (ji) -entry of A .
- The row space of A is $R(A^T)$, the column space of A^T . It is also sometimes called the **coimage** of A .
- The nullspace of A^T is called the **left nullspace** of A . This is also sometimes called the **cokernel** of A .
- **Fundamental Theorem of Linear Algebra, Part 1:** Let A be an $m \times n$ matrix of rank r . Then
 1. $R(A^T)$ = the row space of A . It has dimension r .
 2. $N(A)$ = the nullspace of A . It has dimension $n - r$.

3. $R(A)$ = the column space of A . It has dimension r .
 4. $N(A^T)$ = the left nullspace of A . It has dimension $m - r$.
- Let A be an $m \times n$ matrix with rank r .
 (Existence of Solutions:) The system $Ax = b$ has *at least* one solution x for every b if and only if the columns of A span $\mathbb{R}^m$. If this happens, then $r = m$. In this case there exists an $n \times m$ matrix C that is a **right-inverse** of A : $AC = I_m$. This is possible only if $m \leq n$.
 (Uniqueness of Solutions:) The system $Ax = b$ has *at most* one solution x for every b if and only if the columns of A are linearly independent. If this happens, then $r = n$. In this case there exists an $n \times m$ matrix B that is a **left-inverse** of A : $BA = I_n$. This is possible only if $m \geq n$.
 - If A is an $n \times n$ square matrix of rank $r = n$, then A is **nonsingular** and $B = C = A^{-1}$ and the system $Ax = b$ has a unique solution for every b . In fact, $x = A^{-1}b$ is the unique solution.
 - For an $n \times n$ matrix A , the following are equivalent (some of these terms haven't been defined yet, but they will be):
 1. A is nonsingular.
 2. The rank of A is n .
 3. The columns of A span $\mathbb{R}^n$ (so $Ax = b$ has at least one solution for every b).
 4. The columns of A are linearly independent (so $Ax = \vec{0}$ only has the trivial solution $x = \vec{0}$).
 5. The rows of A span $\mathbb{R}^n$.
 6. The rows of A are linearly independent.
 7. Elimination can be completed: $PA = LDU$ with all $d_i \neq 0$.
 8. There exists a matrix A^{-1} such that $A^{-1}A = AA^{-1} = I_n$.
 9. The determinant of A is not zero: $\det A \neq 0$.
 10. Zero is not an eigenvalue of A .
 11. $A^T A$ is positive definite.

- The **norm** of a vector x is the positive square root of

$$\|x\|^2 = x_1^2 + x_2^2 + \cdots + x_n^2$$

- Two vectors in $\mathbb{R}^n$ are orthogonal if they satisfy the **Pythagorean Theorem**:

$$\|x\|^2 + \|y\|^2 = \|x - y\|^2$$

By the definition of norm given above, this is the same as

$$(x_1^2 + \cdots + x_n^2) + (y_1^2 + \cdots + y_n^2) = (x_1 - y_1)^2 + \cdots + (x_n - y_n)^2$$

Expanding the right-hand side gives

$$(x_1^2 + \cdots + x_n^2) + (y_1^2 + \cdots + y_n^2) = (x_1^2 + \cdots + x_n^2) - 2(x_1y_1 + \cdots + x_ny_n) + (y_1^2 + \cdots + y_n^2)$$

Canceling the terms on both sides and dividing by -2 shows that the Pythagorean Theorem is only satisfied (and so the vectors x and y are orthogonal) when

$$x_1y_1 + \cdots + x_ny_n = 0$$

Please note that this is equivalent to the vector equation

$$x^T y = 0$$

- The **inner product** of two vectors x and y (both in $\mathbb{R}^n$), denoted either by $x \cdot y$ or $\langle x, y \rangle$ is defined to be

$$\langle x, y \rangle = x^T y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

- Two vectors x and y are **orthogonal** if and only if $\langle x, y \rangle = 0$.
- If a set of vectors $v_1, v_2, \dots, v_k$ are mutually orthogonal (so $\langle v_i, v_j \rangle = 0$ for all $i \neq j$) then they are linearly independent.
- Two subspaces W_1 and W_2 of the same vector space V are called **orthogonal subspaces** if every vector w_1 in W_1 is orthogonal to every vector w_2 in W_2 : $\langle w_1, w_2 \rangle = 0$.
- For any $m \times n$ matrix A , the nullspace $N(A)$ and the row space $R(A^T)$ are orthogonal subspaces of $\mathbb{R}^n$. Similarly, the left nullspace $N(A^T)$ and the column space $R(A)$ are orthogonal subspaces of $\mathbb{R}^m$.
- Given a subspace W of a vector space V , the space of all vectors orthogonal to W is called the **orthogonal complement** of W and is denoted $W^\perp$.
- **Fundamental Theorem of Linear Algebra Part 2:** For any $m \times n$ matrix A we have

$$N(A) = (R(A^T))^\perp$$

$$N(A^T) = (R(A))^\perp$$

$$R(A^T) = (N(A))^\perp$$

$$R(A) = (N(A^T))^\perp$$

This is saying that the nullspace and the row space are orthogonal complements, as are the left nullspace and the column space.

This can be visualized in the following diagram (the “four squares” diagram):

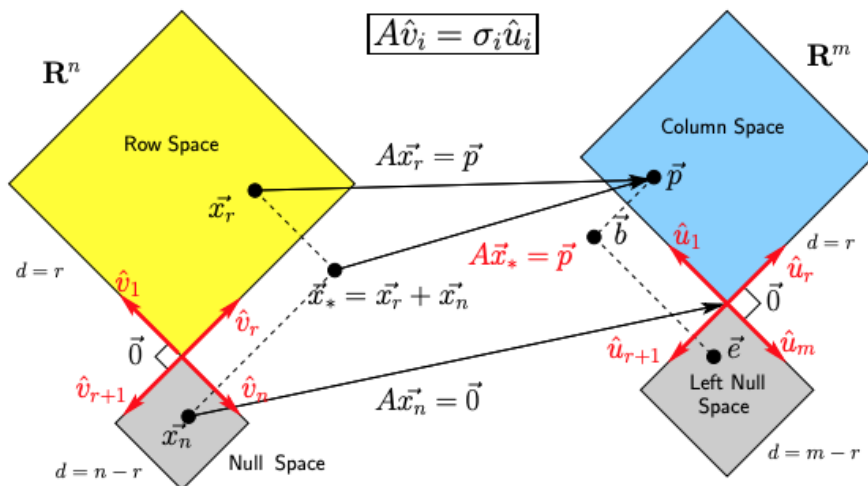


Figure 1: The Fundamental Theorem of Linear Algebra

5 Examples

- Examples of vector spaces.
 - The space $\mathbb{R}^n$. The vectors are column vectors v where each entry in v is a real number. Please note that n can equal ∞ : $\mathbb{R}^\infty$ is the vector space of all infinite-length column vectors. One can think of the elements of $\mathbb{R}^\infty$ as the space of all real-valued sequences.
 - The space $M_{m \times n}(F)$, where F is any field. The vectors are $m \times n$ matrices with all entries from the field, F . When $F = \mathbb{R}$, this is the vector space of all real-valued $m \times n$ matrices. When $F = \mathbb{C}$, this is the space of all complex-valued $m \times n$ matrices. We will work primarily with $M_{m \times n}(\mathbb{R})$.
 - The space $P_n(F)$ of all polynomials of degree less than or equal to n , with coefficients from the field F . The vectors are all polynomials of the form $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, with all $a_i \in F$. For us, F will almost always be $\mathbb{R}$, though it might sometimes be $\mathbb{C}$.
 - General function spaces over some interval $[a, b]$. Depending on the context, the intervals can be open or semi-open (open on one end, closed on the other). For notational simplicity, we will write them as closed intervals here. We allow either $a = -\infty$, $b = \infty$, or both, so we allow infinite intervals.

Examples include (all functions have the stated property on the given interval): $C([a, b])$, the space of all continuous functions; $C^\infty([a, b])$, the space of all **smooth** (i.e. infinitely differentiable) functions; $D([a, b])$, the space of differentiable functions; $I([a, b])$, the space of integrable functions; and so on.

6 Sage Code

6.1 Commands

These are the most common Sage commands that we will use this week. Think of this more as a reference. It is usually easier to understand what the commands do by seeing them used in example code, which is given below.

- `.augment(b)`

This will create the augmented matrix $[A|b]$ by adding the column vector b to the matrix A that the command is appended to. The vector b will either have to be defined prior to this command (the preferable way to do it) or defined directly inside the parentheses.

- `.rank()`

This will calculate the rank of the matrix it is appended to.

- `.pivots()`

This will give a list of the indices of the columns that span the column space. This is more or less the same thing as saying which columns have pivots in the rref, which is more or less the same thing as giving the basic variables. **PLEASE NOTE:** Sage indexes elements in lists starting with 0. This means that you will need to add 1 to each item in the list to correctly give the columns of A .

- `.pivot_rows()`

This will give a list of the indices of the rows that span the row space. Same comment here about the indices starting at 0.

- `.column_space()`

This will give you a basis for the column space, $R(A)$, of the matrix it is appended to. It will also tell you some information about the structure of the subspace, the most important of which will be the rank (which is just the number of vectors in the basis). **PLEASE NOTE:** there are infinitely many different bases, and this command will just give one particular basis. This will usually be different from the basis that you would find if you did this “by hand” using elimination.

- `.row_space()`

This will give you a basis for the row space, $R(A^T)$, of the matrix it is appended to. It will also tell you some information about the structure of the subspace, the most important of which will be the rank (which is just the number of vectors in the basis). **PLEASE NOTE:** there are infinitely many different bases, and this command will just give one particular basis. This will usually be different from the basis that you would find if you did this “by hand” using elimination.

- `.right_kernel()`

This will give you a basis for the nullspace, $N(A)$, of the matrix it is appended to. It will also tell you some information about the structure of the subspace, the most important of which will be the rank (which is just the number of vectors in the basis). **PLEASE NOTE:** there are infinitely many different bases, and this command will just give one particular basis. This will usually be different from the basis that you would find if you did this “by hand” using elimination.

- `.left_kernel()`

This will give you a basis for the left nullspace, $N(A^T)$, of the matrix it is appended to. It will also tell you some information about the structure of the subspace, the most important of which will be the rank (which is just the number of vectors in the basis). **PLEASE NOTE:** there are infinitely many different bases, and this command will just give one particular basis. This will usually be different from the basis that you would find if you did this “by hand” using elimination.

6.2 Example Code

- Okay, this will be a longer example, which will include many of the topics discussed this week and will use lots of Sage, so that’s why it’s included here.

The main object of interest will be the matrix

$$B = \begin{bmatrix} 1 & -3 & -7 & 9 \\ 0 & 1 & 5 & -3 \\ 1 & -2 & -2 & 6 \end{bmatrix}$$

This matrix has size 3×4 , so it maps vectors in $\mathbb{R}^4$ into vectors in $\mathbb{R}^3$. I can’t stress this enough, but **watch the order** here: the first number in the size of the matrix (the number of rows) is the dimension of the range space; the second number in the size of the matrix (the number of columns) is the dimension of the domain space.

We want to understand the nature of the matrix equation $Bx = v$. We’ll start by defining the matrix B and then looking at the augmented matrix $A = [B|v]$, where $v = (a, b, c)^T$. The code below does this in Sage. Please note that to work with the “symbolic vector” (a, b, c) you’ll want to use sympy. See the code below.

```
B = matrix([[1, -3, -7, 9], [0, 1, 5, -3], [1, -2, -2, 6]])
B
```

```
[ 1 -3 -7  9]
[ 0  1  5 -3]
[ 1 -2 -2  6]
```

```
import sympy as sy
sy.init_printing()
a, b, c = sy.symbols('a b c', real = True, constant = True)
A = sy.Matrix([[1, -3, -7, 9, a], [0, 1, 5, -3, b], [1, -2, -2, 6, c]])
A, A.echelon_form()
```

$$\left(\begin{bmatrix} 1 & -3 & -7 & 9 & a \\ 0 & 1 & 5 & -3 & b \\ 1 & -2 & -2 & 6 & c \end{bmatrix}, \begin{bmatrix} 1 & -3 & -7 & 9 & a \\ 0 & 1 & 5 & -3 & b \\ 0 & 0 & 0 & 0 & -a - b + c \end{bmatrix} \right)$$

Okay, what is this telling us? The last line of the echelon form of A is giving us the **compatibility condition** that must be satisfied by the vector b : it is saying that $-a - b + c = 0$, which is equivalent to $c = a + b$. This means that every vector that lies in the column space of B will have the third component equal to the sum of the first two components.

Let's find the column space and the kernel of B . The code below does this.

```
B.column_space()
```

```
Free module of degree 3 and rank 2 over Integer Ring
Echelon basis matrix:
[1 0 1]
[0 1 1]
```

```
B.right_kernel()
```

```
Free module of degree 4 and rank 2 over Integer Ring
Echelon basis matrix:
[ 8  2 -1 -1]
[ 0  3  0  1]
```

This might not be obvious, but if you look at the basis vectors for the column space, they are

$$c_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

This means that any vector in the column space can be written as $ar_1 + br_2$ for constants a, b . But if you look at what this means, we have that for any $v \in R(B)$ we must have

$$v = ac_1 + bc_2 = a \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \\ a + b \end{bmatrix}$$

This is just what we said a few paragraphs ago. Okay, now we want to look at the kernel of B . We see from the above Sage code (please note that you need to use the `.right_kernel()` command for this) that it has a basis given by

$$n_1 = \begin{bmatrix} 8 \\ 2 \\ -1 \\ -1 \end{bmatrix}, \quad n_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \\ 1 \end{bmatrix}$$

This means that any linear combination $k_1 n_1 + k_2 n_2$ will get mapped to $\vec{0} \in \mathbb{R}^3$. The Sage code below shows an example of this.

```
k1 = -1
k2 = 3
n1 = matrix(4,1,[8, 2, -1, -1])
n2 = matrix(4,1,[0, 3, 0, 1])
B*n1, B*n2, B*(k1*n1+k2*n2)

(
[0] [0] [0]
[0] [0] [0]
[0], [0], [0]
)
```

Now let's find a basis for the row space of B . The Sage code below does this.

```
B.row_space()

Free module of degree 4 and rank 2 over Integer Ring
Echelon basis matrix:
[ 1  0  8  0]
[ 0  1  5 -3]
```

We see that a basis is given by

$$r_1 = \begin{bmatrix} 1 \\ 0 \\ 8 \\ 0 \end{bmatrix}, \quad r_2 = \begin{bmatrix} 0 \\ 1 \\ 5 \\ -3 \end{bmatrix}$$

So, we have a basis for the row space of B and the nullspace of B , both of which live in $\mathbb{R}^4$. In fact, they are orthogonal complements and so the vectors $\{r_1, r_2, n_1, n_2\}$ form a basis for $\mathbb{R}^4$. What's more, $\{r_1, r_2\}$ is orthogonal to $\{n_1, n_2\}$ (check this!). This means that any vector x in $\mathbb{R}^4$ can be written in a unique way as a linear combination of these four basis vectors: $x = k_1 r_1 + k_2 r_2 + k_3 n_1 + k_4 n_2$. Let's look at a particular example: $x = (7, -2, 1, 3)^T$. To find the values of the k 's, we can solve a system of equations for them (or have Sage do it for us). The code for that is given below.

```
C = matrix([[1, 0, 8, 0], [0, 1, 2, 3], [8, 5, -1, 0], [0, 3, -1, 1]])
x = matrix(4,1,[7, -2, 1, 3])
C.solve_right(x)

[-1]
[ 2]
[ 1]
[-2]
```

The matrix C has the vectors r_1, r_2, n_1, n_2 as columns and we're using the `.solve_right()` command with the vector x . The solution $(-1, 2, 1, 2)^T$ is telling us that $x = -r_1 + 2r_2 + n_1 - 2n_2$. What does this all mean? It means that x can be written as a linear combination of a vector, x_r that lives in the row space of B and a second vector, x_n , that lives in the nullspace of B .

$$x = \begin{bmatrix} 7 \\ -2 \\ 1 \\ 3 \end{bmatrix} = x_r + x_n = (-r_1 + 2r_2) + (n_1 - 2n_2) = \begin{bmatrix} -1 \\ 2 \\ 2 \\ 6 \end{bmatrix} + \begin{bmatrix} 8 \\ -4 \\ -1 \\ -3 \end{bmatrix}$$

Putting it all together, since matrix multiplication is linear, we have that

$$Bx = B(x_r + x_n) = Bx_r + Bx_n = Bx_r$$

because $Bx_n = \vec{0}$ since x_n is in the nullspace of B . Let's check this.

```
x = matrix(4,1,[7, -2, 1, 3])
xr = matrix(4,1,[-1,2,2,6])
xn = matrix(4,1,[8,-4,-1,-3])
B*x, B*xr, B*xn
```

```
(
 [33] [33] [0]
 [-6] [-6] [0]
 [27], [27], [0]
)
```

This is exactly what we wanted! As an added bonus, and connecting things back to the beginning, notice that Bx is of the form $(a, b, a + b)^T = (33, -6, 33 + (-6))^T = (33, -6, 27)^T$, like we said every vector in the column space must be.

The only thing that we haven't talked about is the left nullspace, which is a subspace of $\mathbb{R}^3$. Let's do that now. Using Sage (you need to use the `left_kernel()` command here), we find that

```
B.left_kernel()
```

```
Free module of degree 3 and rank 1 over Integer Ring
Echelon basis matrix:
[ 1  1 -1]
```

This means that the left nullspace is one-dimensional and has a basis given by $(1, 1, -1)^T$. The columns space and the left nullspace are orthogonal complements and taken together they give all of $\mathbb{R}^3$. This means that a basis for $\mathbb{R}^3$ is given by $\{c_1, c_2, l_1\}$, where

$$c_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad l_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

We won't do it all here, but you can do everything that we just did with $B : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, but with $B^T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$, where B^T is the transpose of B .

- Now, here's the hard truth about all of this. If you do it by hand you'll get completely different answers, but they will still be the right answers! That's one of the incredible (and incredibly frustrating) things about Linear Algebra! Any vector space (or subspace) has infinitely many different bases. Sage is only going to give you one basis. If you do things by hand, you'll usually get a different basis. But they describe the same vector space (or subspace). The easiest way to do this stuff by hand is to work with the echelon form (or the rref) form of the matrix. Here's how you do it.

Given an $m \times n$ matrix B with echelon form U , to find a basis for

- Column Space $R(B)$: Choose the r columns of B in which the pivots appear in U .
- Nullspace $N(B)$: Write the general solution to $Bx = \vec{0}$ as a linear combination of the $n - r$ basis vectors whose coefficients are the free variables. This is easiest to do by actually solving $Ux = \vec{0}$, since the nullspace of U is the same as the nullspace of B .
- Row Space $R(B^T)$: Choose the r nonzero rows of U .
- Left Nullspace $N(B^T)$: Write the general solution to the **adjoint system** $B^T y = \vec{0}$ as a linear combination of the $m - r$ basis vectors whose coefficients are the free variables. Again, this is usually easiest if you instead solve $\tilde{U}y = \vec{0}$, where $\tilde{U}$ is the echelon form of B^T . **PLEASE NOTE:** $\tilde{U} \neq U^T$, so you will need to transpose the matrix then find the echelon form of the transpose.

Here's the output of Sage giving the matrix B and the echelon form U :

```
B, B.echelon_form()

(
 [ 1 -3 -7  9]  [ 1  0  8  0]
 [ 0  1  5 -3]  [ 0  1  5 -3]
 [ 1 -2 -2  6], [ 0  0  0  0]
)
```

So, for example, since we know that the pivots of U are in the first and second columns, we can take the first and second columns of B as a basis for the column space:

$$\tilde{c}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \tilde{c}_2 = \begin{bmatrix} -3 \\ 1 \\ -2 \end{bmatrix}$$

For the row space, we can take the two non-zero rows of U as a basis. This gives

$$r_1 = \begin{bmatrix} 1 \\ 0 \\ 8 \\ 0 \end{bmatrix}, \quad r_2 = \begin{bmatrix} 0 \\ 1 \\ 5 \\ -3 \end{bmatrix}$$

which is the same thing that Sage gave with the `.row_space()` command. If you did Gaussian Elimination to find the echelon form, you probably would have gotten

$$\tilde{r}_1 = \begin{bmatrix} 1 \\ -3 \\ -7 \\ 9 \end{bmatrix}$$

Things are a bit more complicated when finding the nullspace, so we'll only do this once, for $N(B)$. You do the exact same thing with B^T to find the left nullspace.

To start, let $x = (x_1, x_2, x_3, x_4)^T$. Then $Ux = \vec{0}$ gives the two equations

$$\begin{array}{ccccccc} x_1 & & & + & 8x_3 & & = & 0 \\ & x_2 & + & 5x_3 & - & 3x_4 & = & 0 \end{array}$$

Write the variables x_1, x_2 (the basic variables) in terms of x_3, x_4 (the free variables) to get

$$\begin{array}{lcl} x_1 & = & -8x_3 \\ x_2 & = & -5x_3 + 3x_4 \end{array}$$

This means that

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -8x_3 \\ -5x_3 + 3x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -8 \\ -5 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 3 \\ 0 \\ 1 \end{bmatrix}$$

The two vectors

$$\tilde{n}_1 = \begin{bmatrix} -8 \\ -5 \\ 1 \\ 0 \end{bmatrix}, \quad \tilde{n}_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \\ 1 \end{bmatrix}$$

are a basis for the nullspace of B . Notice that $\tilde{n}_2 = n_2$ but $\tilde{n}_1 \neq n_1$.

You can check that $\{r_1(\text{or } \tilde{r}_1), r_2\}$ is orthogonal to $\{\tilde{n}_1, \tilde{n}_2\}$.

As you can see, there are some differences when you do things by hand instead of using Sage. You should be able to do them both ways. You won't be able to use a computer on exams, so please make sure that you can do these things by hand and you can explain what you are doing. Most importantly, you need to be able to **use** the vectors that you find and explain what they are doing with respect to the original matrix B and its various subspaces.

- As a final comment, please note that `B.row_space()` is the same as `(B.T).column_space()` and `B.left_kernel()` is the same as `(B.T).right_kernel()`, and so on. It's probably easier to not use the transpose, but that's up to you.

7 Additional Comments

- If you're feeling a little bit overwhelmed by all of this, that's okay. There's a lot going on and it will take some time to get used to it. Don't be surprised if it takes you a while to "get it," and even once you feel comfortable you'll probably get confused all over again the next time you look at it. The best thing to go is start at the beginning and go through all of the individual steps one at a time. It's also a good idea to write things down and keep track of everything.
- A very useful thing to do is keep track of the size of the spaces involved and where the various subspaces live (and their size). The "four-squares" diagram can be very helpful here.

- This has already been mentioned a couple of times, but it's going to be mentioned again. Any nontrivial vector space will have infinitely many different bases that all generate the same vector space. This can be unbelievably confusing. In general, all you need is one basis in order to do explicit calculations. If you think that you have a basis and Sage (or someone else, like me, for example) has a different basis, there's a very high likelihood that both are correct. Don't let this throw you off. If you can write the other basis in terms of your basis, then you're good to go. Again, the important things are that you actually have a basis (and know how to tell that it is a basis) and that you know what to do with it once you have it. These things take time and practice, so make sure you take your time with all of this and you get lots of practice.

8 Reading Notes

- Olver 2.1 - Real Vector Spaces
- Olver 2.2 - Subspaces
- Olver 2.3 - Span and Linear Independence
- Olver 2.4 - Basis and Dimension
- Olver 2.5 - The Fundamental Matrix Subspaces
- Olver 7.2 - Linear Transformations, Change of Basis

9 Homework Exercises

Please do **ALL** exercises. Not all of them will be graded for correctness, but they will all be checked for completion. Failure to do all of them will lower the completion portion of the Homework score.

Please write your solutions up clearly and cleanly, using full English sentences. Try not to have a bunch of arrows pointing from one thing to another or a bunch of things crossed-out and a bunch of other things circled, etc. Presentation matters.

1. Verify that the nullspace of A is a subspace. To do this, show that the solutions to any homogeneous equation $Ax = \vec{0}$ are closed under addition and scalar multiplication.
2. Let

$$M = \begin{bmatrix} 0 & 0 & 0 & 1 & 2 \\ 1 & -5 & -1 & 0 & -3 \\ -2 & 10 & 3 & 5 & 17 \\ -2 & 10 & 3 & 2 & 11 \end{bmatrix}$$

be the augmented matrix of the linear system given by the matrix equation $Ax = b$, and let

$$R = \begin{bmatrix} 1 & -5 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

be the rref of M .

- (a) How many equations and how many variables does the linear system have?

- (b) Is the linear system consistent?
- (c) What are the pivot positions?
- (d) What are the basic and free variables?
- (e) Describe the solutions.
- (f) Describe the solutions to the homogeneous equation $Ax = \vec{0}$.

3. Let

$$A = \begin{bmatrix} 1 & 4 & 3 & -5 & -11 & 10 \\ 0 & 0 & 1 & -1 & -3 & 3 \\ -1 & -4 & -5 & 7 & 17 & -16 \\ 1 & 4 & 2 & -4 & -8 & 7 \end{bmatrix}$$

and let

$$R = \begin{bmatrix} 1 & 4 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & -1 & -3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

be the rref of A .

- (a) What are the pivot columns of A ?
- (b) What two vectors are guaranteed to span the column space of A ?
- (c) All of the other column vectors in A can be written in terms of the vectors from the previous part. Write these columns (the four ones different from the answer you gave to the previous part) of A as linear combinations of the vectors from part (b).

4. Let the matrix A be given by

$$A = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 0 & 2 & 8 & 0 \end{bmatrix}$$

- (a) Find the echelon form of A .
- (b) What are the basic variables and free variables?
- (c) Find the general solution to $Ax = \vec{0}$.
- (d) Let $b = (b_1, b_2)^T$. Find the conditions on b_1 and b_2 for $Ax = b$ to be consistent.
- (e) Find the general solution to $Ax = b$.
- (f) What is the rank of A ?

5. Find the value of c which makes it possible to solve

$$\begin{array}{rrcrcl} x_1 & + & x_2 & + & 2x_3 & = & 2 \\ 2x_1 & + & 3x_2 & - & x_3 & = & 5 \\ 3x_1 & + & 4x_2 & + & x_3 & = & c \end{array}$$

6. Consider the matrix A with its rref R given below:

$$A = \begin{bmatrix} 1 & -1 & 2 & 6 & -3 \\ 2 & -1 & 0 & 7 & 10 \\ -2 & 3 & -7 & -15 & 17 \\ 2 & -2 & 2 & 8 & 5 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 & 0 & 5 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) Find a basis for the nullspace of A . What is the dimension of the nullspace?
- (b) Which columns of A form a basis for the column space of A . What is the dimension of the column space?
- (c) Write the 4th column of A as a linear combination of the other columns of A .
- (d) Thinking of the matrix A as a linear transformation from $\mathbb{R}^5$ to $\mathbb{R}^4$, is the linear transformation one-to-one? Is it onto? Please justify your answers.

7. Find the dimension and a basis for the fundamental subspaces of

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

- 8. In $\mathbb{R}^3$, find all vectors that are orthogonal to both $(1, 1, 1)$ and $(1, -1, 0)$. Produce from these vectors an orthonormal basis for $\mathbb{R}^3$. (Hint: start by showing that the two vectors are orthogonal. Then consider the vector $x = (x_1, x_2, x_3)$ and assume that it is orthogonal to both of the vectors above. This will give you a system of equations. Solve this system. You now have three mutually orthogonal vectors in $\mathbb{R}^3$. Explain why you know that these three vectors form a basis. Lastly, normalize the vectors to form an orthonormal basis.)
- 9. Explain how you know that the i th row of an invertible matrix B is orthogonal to the j th column of B^{-1} if $i \neq j$.
- 10. Find a basis for the nullspace of

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{bmatrix}$$

and verify that it is orthogonal to the row space. Given $x = (3, 3, 3)$ split it into a row space component x_r and a nullspace component x_n .